



Dr. Santos Kumar Das
Department of ECE
National Institute of Technology Rourkela,
Odisha- 769008, India
Ph: 9437940105 (M), 0661-2462466 (O)
Email: dassk@nitrkl.ac.in

28th August 2024

TO WHOM IT MAY CONCERN

I am pleased to write this letter of recommendation for Ansuman Panigrahi from Odisha University of Technology & Research. Ansuman completed an internship under my guidance at the National Institute of Technology, Rourkela, from May 15, 2024, to July 10, 2024.

During his internship, Ansuman focused primarily on communication protocols, gaining rapid and in-depth exposure to various technologies, including LoRa, Free Space Optics (FSO), GPS, and GSM modules. He quickly grasped and implemented these protocols in practical applications, demonstrating his adaptability and technical proficiency. His contributions significantly enhanced our communication systems projects.

In addition to his expertise in communication protocols, Ansuman developed a comprehensive understanding of web development concepts. He displayed a thorough grasp of the principles, applications, and techniques of web development using protocols such as HTTPS and languages like JavaScript, Node.js, Express.js, HTML, and CSS.

Ansuman was also involved in a cutting-edge project that integrated ESP32 to develop a smart traffic system with various advantages over traditional traffic systems. The system features two modes for optimal switching and control of traffic and can be managed through a web interface, allowing modular settings adjustments.

Furthermore, Ansuman demonstrated proficiency with the ESP32 and Arduino IDE, exploring advanced techniques and testing an IoT development kit equipped with IMU sensors, barometric sensors, OLED displays, FSO modules, GSM modules, DHT11 sensors, and more. He also gained experience in I2C and serial interfacing.

Additionally, Ansuman tested the NVIDIA Jetson TX2 and acquired knowledge of Linux OS, further broadening his technical expertise.

Throughout these projects, Ansuman consistently exhibited qualities essential for success in engineering and technology. Apart from his technical skills, he demonstrated high levels of professionalism and teamwork. His collaborative spirit and clear communication greatly contributed to a productive working environment.

In conclusion, I highly recommend Ansuman for any opportunity that allows him to further develop and apply his skills. He has proven to be a valuable asset in any technical environment, and I am confident that he will continue to excel in his future endeavours.

Thanking you.

Yours truly,

Dr. Santos Kumar Das
Associate Professor
Dept. of Electronics & Communication Engg.
National Institute of Technology Rourkela
Odisha - 769008, India
Tel: +916612462166 (O), M-9437940105
dassk@nitrkl.ac.in